

Relative Performance Feedback and Redistribution Preferences: Evidence from an Online Experiment*

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Abstract

This paper examines how relative performance feedback affects redistribution preferences and beliefs about effort versus luck in generating inequality. In an online experiment, participants complete a code-recognition task and are randomly assigned to one of three conditions: no feedback, told they performed *above* a randomly matched peer, or told they performed *below*. They then act as impartial spectators and redistribute a fixed sum between two workers whose earnings reflect their performance. Participants told they outperformed their peer score 1.21 points higher on an effort-attribution scale (0–10) than those told they underperformed ($p < 0.05$) and redistribute \$0.52–0.55 less to the lower-earning worker ($p < 0.10$). The no-feedback group sits between the two treatment arms; neither its belief score nor its redistribution differs significantly from either treated group. A mediation analysis shows that roughly 47 percent of the redistribution effect is explained by the shift in effort–luck beliefs, consistent with self-serving attribution as the channel linking feedback to redistribution. As a novel design element, participants report their expected relative performance before receiving the feedback signal; we find that 84 percent predicted to outperform, reflecting a strong optimism bias that constrains identification of the surprise channel in this sample.

Keywords: redistribution, relative performance feedback, self-serving attribution, fairness,

*This experiment builds on the design of Almås et al. (2020) and is inspired by the research question of Cassar and Klein (2019).

online experiment

JEL codes: C91, D31, D63, D91

1 Introduction

Free-market economies inherently generate unequal outcomes, and while modern democratic societies rely on redistribution to mitigate these disparities, there is persistent disagreement over which inequalities are considered legitimate (Alesina and Giuliano, 2011; Piketty, 2014). A central dimension of this debate concerns the *source* of inequality: outcomes attributed to effort are typically viewed as more deserving than those arising from luck. This intuition is formalised in the accountability framework, which models redistributive preferences as judgments about deservingness and predicts lower redistribution when inequality reflects effort rather than chance (Konow, 2000; Cappelen et al., 2007). The framework finds broad empirical support: beliefs about the determinants of income are consistently among the strongest predictors of redistribution preferences (Piketty, 1995; Alesina and Angeletos, 2005; Bénabou and Tirole, 2006).

Yet a key difficulty in this debate is that, in practice, the source of inequality is rarely clear-cut. Individual success reflects a mix of effort, innate ability, family background, and circumstance, and it is often hard to separate these contributions, even for the person directly involved (Piketty, 2014). When effort and luck cannot be cleanly distinguished, beliefs about which is responsible do not simply reflect the available evidence; they are also shaped by personal experience and motivated reasoning (Kunda, 1990). Since these beliefs drive redistribution preferences, whatever shapes them also shapes redistribution demand.

Deffains et al. (2016) argue that the experience of success and failure shapes redistribution preferences through the self-serving bias: individuals who succeed at a task attribute their outcome to effort and favour less redistribution, while those who fail attribute their outcome to circumstance and shift toward more. Since wealthier individuals have typically experienced more success, this bias systematically pushes them to view inequality as earned rather than lucky, creating a divide between rich and poor not just in material interests but also in fairness views. As Deffains et al. (2016) put it, wealthy individuals “do not only oppose redistribution because they expect to be net payers, but also because the self-serving bias systematically shifts their fairness principles.” Identifying this channel outside a controlled setting is difficult, however: in observational data one cannot tell whether successful individuals attribute outcomes to effort because of cognitive bias or because they genuinely worked harder. Isolating the self-serving bias therefore requires exogenous variation in comparative signals, holding actual performance constant while shifting only what individuals learn about their relative standing. This paper asks whether such a signal — being told one outperformed or underperformed a peer — is enough to shift beliefs about the role of effort in generating inequality and, through that channel, redistribution preferences.

This paper provides experimental evidence on this mechanism. Participants complete a performance task and are randomly assigned to one of three conditions: (i) *no feedback* about relative standing; (ii) told they performed *above* a randomly matched peer (*above* group); or (iii) told they performed *below* a matched peer (*below* group). Subsequently, all participants act as impartial spectators and decide how to distribute a fixed monetary sum between two workers whose inequality arose from differential performance.

The key findings are four. First, participants told they outperformed a peer redistribute less in the merit scenario: \$1.91 on average, versus \$2.44 in the below group and \$2.12 in the no-feedback group. The above–below gap of \$0.52–0.55 is marginally significant ($p < 0.10$) across both specifications. Second, the same group scores 1.21 points higher on an effort-attribution scale (0–10) than the below group ($p < 0.05$), consistent with self-serving belief updating; the joint test across all three groups also rejects equality of means ($p = 0.041$, ANOVA). A mediation analysis shows that roughly 47 percent of the redistribution gap is accounted for by the shift in effort–luck beliefs, providing direct evidence that self-serving attribution is the operative channel linking the comparative signal to redistribution. Third, the no-feedback group anchors in the middle of both outcomes. We include this arm specifically to ask whether the *information content* of the comparative signal — not task completion alone — is what drives belief updating. Its intermediate position is consistent with that interpretation: both comparative signals push beliefs and redistribution away from an uninformed baseline in opposite directions. Linear hypothesis tests cannot reject equality of the below and no-feedback groups ($p = 0.26$ for redistribution, $p = 0.12$ for beliefs), and neither treated group individually differs from the no-feedback anchor at conventional significance levels; the statistically identifiable contrast is the above–below gap. Fourth, by classifying participants according to whether their feedback confirmed or contradicted their pre-feedback prediction, we identify a *confirmation* channel. Among participants who expected to outperform their peer, those whose expectation was confirmed (*positively reassured*: told above, predicted above; $n = 42$) score 1.15 points higher on effort attribution and redistribute \$0.63 less than those whose expectation was contradicted (*negatively surprised*: told below, predicted above; $n = 37$), with both contrasts significant at the 5 percent level. This is the sharpest contrast in the data: confirmation of a prior belief in one’s own superiority triggers the strongest merit-oriented response, consistent with motivated reasoning amplifying the self-serving attribution effect when feedback reinforces rather than challenges an existing self-view.

These results are consistent with the self-serving attribution account developed by Defains et al. (2016): individuals who perform well relative to others update their beliefs about the role of effort in generating inequality and become less redistributive. Our design extends

their framework by randomly assigning the comparative signal, providing clean causal identification and isolating the relative — rather than absolute — performance channel. The paper most directly relates to Almås et al. (2020), from whose spectator design we adapt the merit redistribution task, and to Cassar and Klein (2019), whose focus on how performance comparisons shape distributive preferences motivates our research question.

A second contribution is a novel design element: before receiving the feedback signal, participants report whether they expect to outperform their matched peer. This allows us to cross feedback content (above, below, none) with prior beliefs (optimistic, pessimistic), yielding six mutually exclusive feedback types. The most informative contrast — and the paper’s sharpest result — is between *positively reassured* participants (told above, predicted above; $n = 42$) and *negatively surprised* participants (told below, predicted above; $n = 37$). Both groups shared identical optimistic priors, but their feedback either confirmed or contradicted those beliefs. The confirmed group scores 1.15 points higher on effort attribution ($p = 0.037$) and redistributes \$0.63 less ($p = 0.042$), consistent with motivated reasoning reinforcing positive self-views when feedback aligns with them. The main limitation is that near-universal overconfidence (84% predicted above) concentrates the positive-surprise cell at just $n = 5$, preventing estimation of the symmetric case and limiting power for the within-arm comparisons more broadly. A design with more balanced prior expectations would be needed to fully map the interaction between prediction direction and feedback valence.

The rest of the paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the experimental design. Section 4 presents the results. Section 5 concludes.

2 Related Literature

This paper contributes to three strands of literature.

Beliefs about inequality and redistribution. A key insight from Piketty (1995) is that beliefs about social mobility and the role of effort versus luck determine preferences for redistribution in equilibrium. Alesina and Angeletos (2005) provide empirical evidence: Americans attribute poverty more to individual failure than Europeans, which partly explains the cross-country gap in redistribution. Bénabou and Tirole (2006) model how self-serving motivated reasoning sustains two equilibria — a “belief in a just world” and a more luck-oriented view — with different steady-state redistribution levels. Our experiment directly tests the mechanism by which an informational shock (relative performance feedback) updates beliefs and behaviour.

Impartial spectator and fairness views. Cappelen et al. (2007) and Cappelen et al. (2013) use spectator designs to classify subjects into fairness types (libertarian, meritocratic, egalitarian) and show that a majority of individuals prefer some redistribution away from inequality that arises from luck rather than effort. We adapt their spectator design to a merit scenario and ask whether third-party redistribution decisions are affected by information about the spectator’s own relative performance — a test of whether the spectator role is truly impartial.

Self-serving attribution and social comparisons. Self-serving attribution refers to the tendency to attribute successes internally (effort) and failures externally (luck or circumstance) (Miller and Ross, 1975; Mezulis et al., 2004). In an economic context, Babcock and Loewenstein (1995) document that parties in bargaining inflate self-assessments of a fair outcome in a self-serving direction. Closer to our design, Deffains et al. (2016) show that individuals who experience unfair treatment become more supportive of redistribution. Cassar and Klein (2019) show that framing one’s own performance as a failure (relative to a counterfactual) shapes distributive preferences; our design extends this insight by using randomly assigned comparative feedback, allowing clean causal identification. We add to this literature by focusing on the *comparative* dimension: the same absolute performance level produces different attribution and redistribution depending on whether one learns one stands above or below a peer. Beyond the main treatment contrast, our pre-feedback elicitation design reveals a *confirmation* channel: the effect is largest for participants whose prior optimism was corroborated by positive feedback, consistent with motivated reasoning models in which updating is asymmetric — reinforcing positive self-views more readily than revising them downward (Bénabou and Tirole, 2006; Kunda, 1990).

3 Experimental Design

The experiment was conducted online using JATOS and hosted the free, ESCoP-sponsored server Mindprobe. The whole design builds on the spectator design of Almås et al. (2020). It involved two distinct roles: *workers* and *spectators*.

Workers ($n = 2$) were recruited through Prolific and completed the code-recognition task under real monetary incentives. They were paid a base fee of \$4 plus a performance-based bonus of up to \$8, for a maximum possible payout of \$12 each. They were informed in advance that their earnings would initially depend on their relative performance against each other—the higher-scoring worker would receive \$8 while the other would receive \$0—and that a third party (a spectator) would subsequently decide whether and how much to

redistribute between them.

Spectators ($n = 134$) were recruited through personal networks (friends, family, and colleagues from two workplaces), completed the full experiment online in approximately 10 minutes, and received no monetary compensation. The experiment consisted of two parts: a code-recognition task and a redistribution decision task in which they acted as impartial spectators. Spectators were informed before making their redistribution decisions that the workers were real individuals who had completed the same task and would receive actual monetary payouts determined by one randomly selected spectator’s allocation. The impartial-spectator design thus removes material self-interest on the spectator side while keeping real stakes for the workers, ensuring that decisions have genuine consequences and giving the design greater external validity.

3.1 Part 1: Code-Recognition Task

Participants (both workers and spectators) completed a code-recognition task consisting of one practice round of 20 seconds followed by five scored rounds of 45 seconds each. At the start of each round, a three-digit target number (drawn from 100–999) was displayed at the top of the screen. Below it, a grid of three-digit numbers appeared; approximately 15 percent of the codes in the grid matched the target (around 48 instances per round), and the rest were distractors. Participants had to identify and check every instance of the target as quickly as possible. The grid was generated randomly at the start of each round, so the target and distractor codes changed every 45 seconds. To ensure a consistent interface experience, the grid dimensions were adapted to screen size: desktop users saw a 20×16 grid (320 codes), while mobile users saw a 7×46 grid (322 codes), with the same number of target instances in both cases. Participants were explicitly told about the scoring system: they earned 1 point for each correct identification, lost 1 point for each false alarm, and received no penalty for a missed target. The maximum possible score across the five rounds was around 200 points; sample means range from 90 to 95 points across treatment groups (Table 1).¹

Just after finishing, participants answered three pre-feedback self-report questions: (i) effort exerted (slider, 0–10); (ii) guessed score (numeric, range -200 to 200); and (iii) relative performance prediction (“I would perform better than or the same as the other participant” vs. “I would perform worse”).

¹This marked the end of the task for the two workers. Before exiting, they were reminded that one spectator, chosen at random, would decide how to split their earnings, and that this decision would be paid out as their bonus. Spectators continued to Part 2, as described below.

3.2 Treatments: Relative Performance Feedback

After the self-report, participants were randomly assigned to one of three feedback conditions.

1. **No feedback.** Participants were informed they could proceed to Part 2 without seeing their score or any comparison information.
2. **Above.** Participants were told: “You performed above your randomly matched peer.”
3. **Below.** Participants were told: “You performed below your randomly matched peer.”

The feedback signal was generated through a dynamic matching procedure. Before the main data collection, ten pilot participants completed the task, providing an initial distribution of scores and establishing a current best and worst performer. As each subsequent spectator completed the task, those randomly assigned to the *above* condition were matched with the current lowest scorer in the spectator pool; those assigned to the *below* condition were matched with the current highest scorer. Reference scores were updated continuously as the experiment progressed. Note that spectators were compared with other spectators in this pool and the matching was indeed random. This design guarantees genuine, non-deceptive feedback: every participant in the above condition truly outperformed their assigned reference peer, and every participant in the below condition truly underperformed theirs. Participants who could not receive their assigned treatment — above-condition participants who scored below the current worst, or below-condition participants who scored above the current best — were excluded from the analysis. At debriefing, participants were shown their actual score and informed of the matching procedure, so they could understand why they had received the message they did.

3.3 Part 2: Distribution Decisions

After the (non)feedback screen, all participants acted as impartial spectators and made a redistribution decision adapted from Almås et al. (2020).

Redistributive Decision in a Merit Scenario

Participants read that two real participants — the two workers who had previously completed the same task — were identified as Worker A and Worker B. Worker A had received a larger initial payment because they performed better; Worker B received nothing because they performed worse. Participants were asked to decide whether and how to redistribute the given endowment between the two workers. The decision was presented as a set of discrete options covering the full range from keeping full inequality to reversing it.

After the redistribution decision, participants completed a short demographics survey (age, gender, and education) followed by the key belief measure of this study. They were asked to respond to the following question using a slider:

“To what extent do you think the initial income (before any redistribution) of the participants in the previous task was determined by effort versus luck?”

The slider ranged from 0 (“Completely determined by luck”) to 10 (“Completely determined by effort”). This variable (`effort_luck`) is the hypothesised mediator between relative performance feedback and redistribution preferences.

3.4 Key Outcome and Mediator Variables

- **Redistribution (merit):** USD allocated to Worker B in the merit scenario. Primary outcome variable.
- **Effort–luck belief (`effort_luck`):** Score on the 0–10 slider described above. Hypothesised mediator.

3.5 Sample

Spectators were recruited through personal networks: friends, family, and employees at two workplaces. The analysis sample consists of $N = 145$ spectators after excluding non-engagers (zero score and zero clicks). The sample is predominantly Colombian (76.5%), with the remainder drawn from other countries, primarily in Europe. As this is a convenience sample, findings should be interpreted with caution regarding external validity and may not generalise to other populations in terms of magnitudes; treatment assignment was, however, random within this sample, so internal validity is preserved.

Table 1 presents descriptive statistics by treatment group. Groups are balanced on all pre-determined covariates: age ($p = 0.94$), gender ($p = 0.57$), task score ($p = 0.71$), nationality ($p = 0.32$), and education ($p = 0.52$), confirming that randomisation was successful. The nationality variable shows the above group with a slightly higher share of Colombians (83%) compared to the no-feedback (70%) and below (77%) groups, though this difference is comfortably non-significant ($p = 0.32$) and attributable to chance given the small sample size.

The effort–luck belief score differs significantly across groups ($p = 0.041$), which is itself a key outcome; importantly, this measure is elicited *after* the feedback treatment, so it cannot confound treatment assignment.

Table 1: Descriptive Statistics by Treatment Group

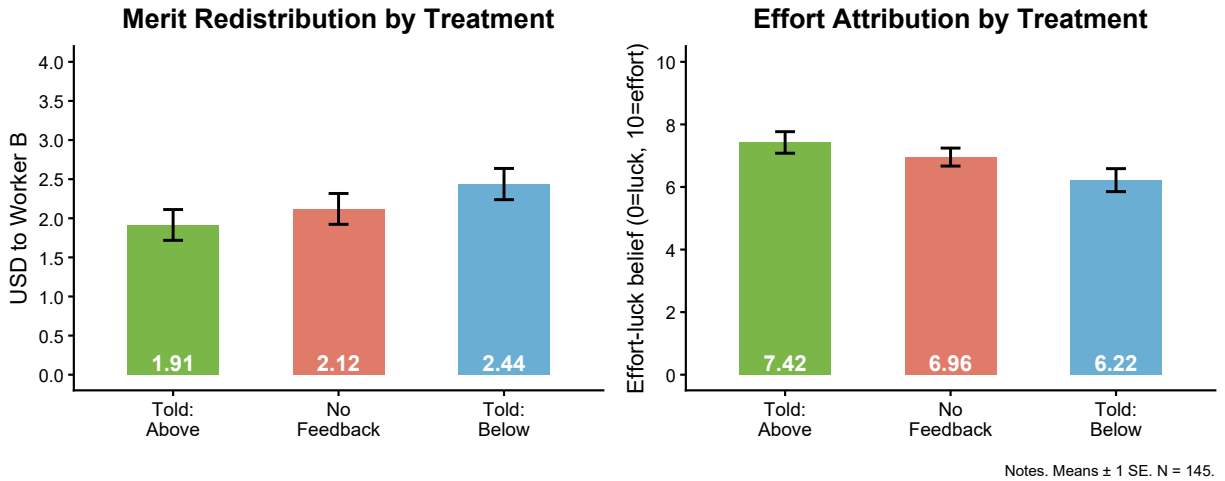
	Told: Above	No Feedback	Told: Below	<i>p</i> -value
Redistribution – merit (\$)	1.91 (1.35)	2.12 (1.39)	2.44 (1.38)	0.179
Effort–luck belief (0–10)	7.42 (2.36)	6.96 (2.03)	6.22 (2.55)	0.041
Task score	94.53 (27.81)	92.50 (26.37)	89.85 (27.89)	0.706
Age	34.77 (13.69)	35.84 (14.97)	35.21 (14.93)	0.935
Female (%)	61.70 (49.14)	62.00 (49.03)	70.83 (45.93)	0.567
Colombian (%)	82.98 (37.99)	70.00 (46.29)	77.08 (42.47)	0.319
Education (%)				0.519
Less than bachelor	23.40 (42.80)	18.00 (38.81)	10.42 (30.87)	
Bachelor’s	48.94 (50.53)	50.00 (50.51)	60.42 (49.42)	
Master’s or above	27.66 (45.22)	32.00 (47.12)	29.17 (45.93)	
<i>N</i>	47	50	48	145

Mean (SD) for continuous variables; mean $\times$ 100 (SD $\times$ 100) for binary.

p-values from one-way ANOVA (continuous) or chi-squared test (binary). Redistribution in USD.

4 Results

Figure 1: Effort Attribution and Merit Redistribution by Treatment Group



Notes. Left panel: mean effort–luck belief (0 = 100% luck; 10 = 100% effort), elicited after the redistribution decision. Right panel: mean USD allocated to the lower-earning worker (merit scenario). Bar heights are group means; whiskers are ± 1 SE. $N = 145$.

Figure 1 gives an overview of both main outcomes across the three treatment arms.

Among the 145 spectators in the sample, 47 were told they outperformed their matched peer (above), 48 were told they underperformed (below), and 50 received no feedback. The left panel shows mean effort–luck belief and the right panel shows mean redistribution to the lower-earning worker; both outcomes display the same monotone gradient. The above group attributes the most to effort and redistributes the least; the below group is the most luck-oriented and the most redistributive; and the no-feedback group sits between the two. The most salient visual feature in both panels is the above–below gap: the distance between the two treated groups is roughly twice the distance of either group from the no-feedback anchor. Neither treated group individually differs from the no-feedback anchor at conventional significance levels; **AQUI TODAVÍA NO SÉ SI PONER ALGO MÁS O NO. TIPO, ESTÁ EN LA MITAD, LOS COEFICIENTES TIENEN SENTIDO, PERO NO TENGO SUFICIENTE PODER, Y LUEGO MIRAR EL ANEXO CUANDO EL BASELINE ES NON-FEEDBACK PORQUE CREO QUE SÍ DA SIGNIFICATIVO** the statistically identifiable contrast is the above–below gap, consistent with both signals moving participants symmetrically from a common uninformed reference point.

4.1 Effort–Luck Beliefs

The right panel shows a clear monotone pattern: the above group scores highest ($\overline{EL}_{\text{above}} = 7.42$, $SD = 2.36$), followed by the no-feedback group ($\overline{EL}_{\text{nofb}} = 6.96$, $SD = 2.03$) and the below group ($\overline{EL}_{\text{below}} = 6.22$, $SD = 2.57$). Notably, all three groups score above the scale midpoint of 5, indicating that effort attribution is the modal view regardless of treatment; what the feedback shifts is the *degree* to which participants see the outcome as effort-determined. A one-way ANOVA rejects equality of group means ($p = 0.041$).

Columns 1–2 of Table 2 report OLS regressions of the effort–luck belief on treatment dummies, with the above group as the baseline. Without controls (column 1), the below group scores 1.21 points lower than the above group ($SE = 0.51$, $p < 0.05$). The no-feedback group scores 0.47 points lower ($SE = 0.45$, $p = 0.30$), roughly half the magnitude of the below-group gap and not statistically significant. Taking the no-feedback group as a natural anchor, the point estimates suggest that positive feedback elevates effort attribution by roughly 0.47 points above this anchor, while negative feedback depresses it by approximately 0.74 points below it — both signals moving in the expected direction, though neither individual shift is large enough to be statistically significant on its own. Once controls are added (column 2), both estimates remain stable: the below-group gap widens slightly to 1.30 points ($SE = 0.52$, $p < 0.05$), and the no-feedback coefficient is -0.48 ($SE = 0.45$), confirming that the pattern

does not reflect pre-existing compositional differences across groups. A linear hypothesis test cannot reject that the below and no-feedback coefficients are equal ($F(1, 142) = 2.45$, $p = 0.12$).

These findings are consistent with the motivated-reasoning account of Kunda (1990): positive comparative signals are more readily internalised because they serve the motivated goal of maintaining a positive self-image, while negative signals invite counter-argument and discounting (Mezulis et al., 2004). Because the comparative signal is exogenously assigned and uncorrelated with actual task performance, these belief differences cannot be explained by genuine differences in effort across groups, providing experimental support for the belief-updating channel theorised by Deffains et al. (2016).

4.2 Redistribution in the Merit Scenario

The left panel shows mean redistribution following the same gradient. The above group allocates the least (\$1.91, SD = \$1.35), followed by the no-feedback group (\$2.12, SD = \$1.39) and the below group (\$2.44, SD = \$1.38). The one-way ANOVA yields $p = 0.179$, so initially we cannot reject equality of means across all three groups.

Columns 3–4 of Table 2 report the main OLS regressions ($N = 145$). Without controls (column 3), the below group redistributes \$0.52 more than the above group (SE = \$0.28, $p < 0.10$), while the no-feedback group redistributes \$0.21 more (SE = \$0.28), a difference that is not statistically significant. With the full set of controls (column 4), the below-group estimate is \$0.55 (SE = \$0.31, $p < 0.10$), remaining marginally significant.

The above–below gap of \$0.52–0.55 is stable across specifications and economically meaningful, representing roughly 27–29% of the above-group mean of \$1.91, and remains marginally significant in both columns. A linear hypothesis test cannot reject that the no-feedback and below groups redistribute the same amount ($F(1, 142) = 1.25$, $p = 0.26$).

4.3 The Belief Channel: Mediation Analysis

If self-serving attribution is the mechanism at work, then the treatment effect on redistribution should operate *through* the effort–luck belief. Table 3 tests this by adding the effort–luck belief score as a covariate to the redistribution regression ($N = 145$).

The effort–luck belief score is a strong and highly significant predictor of redistribution: each one-point increase on the 0–10 scale is associated with \$0.20 less redistributed to the lower-earning worker (SE = \$0.05, $p < 0.01$), both with and without additional controls. Once effort–luck belief is included, the treatment coefficients become small and statistically indistinguishable from zero: the below-group coefficient falls from \$0.52 (column 3 of Table 2)

Table 2: Treatment Effects on Effort Attribution and Redistribution

	USD Redistributed		Effort Beliefs	
	(1)	(2)	(3)	(4)
Constant	1.915*** (0.199)	3.461*** (0.837)	7.423*** (0.348)	6.470*** (1.301)
No Feedback	0.205 (0.281)	0.172 (0.290)	-0.467 (0.453)	-0.476 (0.450)
Told: Below	0.523* (0.283)	0.549* (0.305)	-1.205** (0.509)	-1.301** (0.518)
Age		-0.008 (0.012)		0.021 (0.018)
Female		-0.256 (0.237)		0.324 (0.405)
Colombian		-0.402 (0.296)		0.413 (0.533)
Task score		-0.005 (0.006)		-0.007 (0.009)
Bachelor's degree		-0.363 (0.365)		0.369 (0.549)
Master's or above		-0.372 (0.404)		0.461 (0.597)
N	145	145	145	145
R^2	0.024	0.077	0.044	0.110
Demographic controls	No	Yes	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS with HC3-robust SEs in parentheses. Baseline: Told Above. Cols. (1)–(2): USD redistributed to lower-earning worker (merit scenario). Cols. (3)–(4): effort–luck belief (0 = luck, 10 = effort). Controls: age, female indicator, Colombian indicator, task score, and education (Less than bachelor = baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

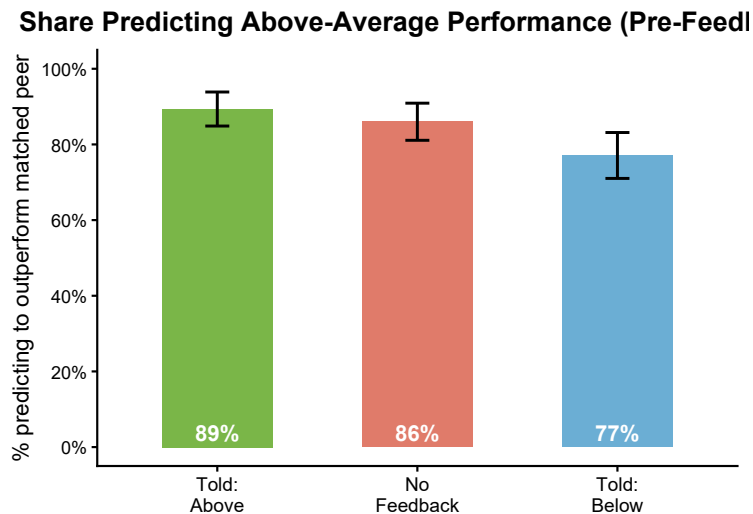
to \$0.28 (SE = \$0.27), a reduction of approximately 47 percent. The no-feedback coefficient falls from \$0.21 to \$0.11 (SE = \$0.27). Both results are consistent with the hypothesis that the feedback signal shifts redistribution *through* beliefs about effort and luck: once we condition on the belief, the direct effect of the signal on redistribution largely disappears.

This pattern is consistent with partial mediation by the effort–luck belief, as theorised by Deffains et al. (2016) and predicted by the accountability framework (Konow, 2000; Cappelen et al., 2007): information about relative standing updates beliefs about the deservingness of inequality, and it is this updated belief — not the signal itself — that drives redistributive behaviour.

4.4 Pre-Feedback Expectations and the Surprise Channel

A novel element of this design is that participants state their expected relative performance *before* receiving the feedback signal. After the code-recognition task but before learning whether they performed above or below their peer, participants were asked to predict whether they would outperform the matched participant. This allows us to classify participants by whether the feedback was *surprising* — that is, whether it contradicted their prior expectation — and to ask whether unexpected feedback triggers stronger belief updating and redistribution responses.

Figure 2: Pre-Feedback Performance Predictions by Treatment Group



Notes. Proportion ± 1 SE. $N = 145$.

Notes. Share of participants predicting they would outperform their matched peer, elicited before the feedback signal was revealed. Bars show group proportions; whiskers are ± 1 SE. $N = 145$.

Figure 2 shows the distribution of pre-feedback predictions by treatment group. The large majority of participants expected to outperform their peer: 89% in the above group,

Table 3: Mediation Analysis

	(1) Merit redistrib. (\$)	(2) Merit redistrib. (\$)
Constant	3.433*** (0.432)	4.717*** (0.796)
No Feedback	0.110 (0.274)	0.080 (0.280)
Told: Below	0.276 (0.274)	0.297 (0.292)
Effort–luck belief	−0.204*** (0.051)	−0.194*** (0.053)
Age		−0.004 (0.011)
Female		−0.193 (0.229)
Colombian		−0.321 (0.291)
Task score		−0.007 (0.005)
Bachelor’s degree		−0.292 (0.339)
Master’s or above		−0.283 (0.378)
<i>N</i>	145	145
<i>R</i> ²	0.140	0.174
Demographic controls	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS with HC3-robust SEs in parentheses. Dependent variable: USD redistributed (merit scenario). Includes effort–luck belief as control to assess mediation; compare treatment coefficients with Table 2 cols. (3)–(4). Baseline: Told Above. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

86% in the no-feedback group, and 77% in the below group (84% overall). This pattern reflects the well-documented optimism bias or “better-than-average” effect (Mezulis et al., 2004): most individuals expect above-average performance even when only half can logically do so.

To isolate the surprise component, Table 4 compares outcomes within each treatment arm by whether the feedback contradicted the participant’s prior expectation. Columns (1) and (3) restrict to the above group and regress each outcome on a dummy for being positively surprised (told above despite predicting below; $n = 5$). Columns (2) and (4) restrict to the below group and regress on a dummy for being negatively surprised (told below despite predicting above; $n = 37$). The constant in each column is the mean for the *not-surprised* sub-group: those whose feedback confirmed their prediction.

None of the surprise coefficients are statistically significant. Among the above group, positively surprised participants score slightly *lower* on effort attribution (-0.50 , $SE = 1.72$) and redistribute slightly *more* ($\$0.32$, $SE = \$0.77$) than the not-surprised above participants — the direction one might expect if an unexpected win feels more “lucky” than a confirmed one, though the estimates are far too imprecise to draw conclusions ($n = 5$). Among the below group, negatively surprised participants score slightly *higher* on effort attribution ($+0.45$, $SE = 0.96$) and redistribute slightly more ($\$0.33$, $SE = \$0.54$) than the not-surprised below participants. A notable pattern is that the not-surprised below group (those who expected to lose and did; $n = 11$) shows the lowest effort attribution of all sub-groups (5.87 on the 0–10 scale), suggesting that individuals who are consistently pessimistic about their own performance also hold stronger luck-oriented beliefs.

Columns (5) and (6) pool all 145 observations and regress effort attribution and redistribution on a six-category factor variable for feedback type, with the *negatively surprised* group (told below, predicted above; $n = 37$) as the baseline. The remaining five categories are: *negatively reassured* (told below, predicted below; $n = 11$), *positively surprised* (told above, predicted below; $n = 5$), *positively reassured* (told above, predicted above; $n = 42$), *no feedback – optimistic* (no feedback, predicted above; $n = 43$), and *no feedback – pessimistic* (no feedback, predicted below; $n = 7$).

Two groups differ significantly from the negatively surprised baseline. First, *positively reassured* participants score 1.15 points higher on effort attribution ($SE = 0.55$, $p = 0.037$) and redistribute $\$0.63$ less ($SE = \$0.31$, $p = 0.042$): those whose optimism was confirmed by the feedback are the most merit-oriented of all groups. Second, *no-feedback pessimists* score 1.46 points higher on effort attribution ($SE = 0.64$, $p = 0.023$) than the negatively surprised baseline, though their redistribution does not differ significantly ($-\$0.37$, $SE = \$0.59$, $p = 0.53$). No other group differs significantly from the negatively surprised baseline in either

outcome. Notably, the *no-feedback optimists* ($n = 43$), who represent the majority of the no-feedback arm, are statistically indistinguishable from the negatively surprised baseline, which explains why the no-feedback treatment coefficient in Table 2 closely mirrors the below coefficient rather than the above.

The divergence between no-feedback pessimists and optimists is intriguing: pessimistic participants who receive no comparative signal still hold effort-oriented beliefs comparable to the positively reassured group, while optimistic no-feedback participants do not. This suggests that pre-existing beliefs about one’s relative standing interact with feedback reception in ways that a single no-feedback dummy cannot capture, and warrants further investigation with larger samples.

Table 4 underscores the main limitation of the within-arm analysis: the R^2 values in columns (1)–(4) are all below 0.01, and the positive-surprise cell is too small ($n = 5$) to yield any useful estimate. The full-sample results in columns (5)–(6) further suggest that the treatment effect documented in Sections 4.1–4.2 is primarily driven by the contrast between those whose optimistic prediction was confirmed (*positively reassured*) and those whose optimistic prediction was contradicted (*negatively surprised*), rather than by the surprise element per se. Identifying the causal role of surprise and the no-feedback heterogeneity would require a sample with more balanced prior expectations and a larger pessimistic no-feedback cell.

5 Conclusion

This paper provides experimental evidence that relative performance feedback shapes beliefs about the role of effort in generating inequality and, through that channel, redistribution preferences. Participants told they outperformed their peer attribute significantly more of the task outcome to effort than those told they underperformed (gap of 1.21 points on a 0–10 scale, $p < 0.05$), and redistribute less in the merit scenario (gap of \$0.52–0.55, $p < 0.10$). A mediation analysis shows that approximately 47 percent of this redistribution gap operates through the shift in effort–luck beliefs, consistent with self-serving attribution as the operative channel. The no-feedback arm anchors in the middle of both outcomes, and neither individual contrast with the treated groups is statistically significant, suggesting that the *direction* of the comparative signal — not mere task completion — is what drives the update; the identifiable effect is the above–below divergence, with both signals displacing beliefs symmetrically from an uninformed baseline.

Beyond the main treatment effect, the pre-feedback performance prediction reveals a confirmation channel. Among participants who expected to outperform their peer, those whose expectation was confirmed (*positively reassured*) score 1.15 points higher on effort

Table 4: Surprise Effects: Within-Arm and Full-Sample Comparisons

	USD Redistributed			Effort Beliefs		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	1.881*** (0.210)	2.182*** (0.486)	2.514*** (0.224)	7.476*** (0.353)	5.873*** (0.865)	6.322*** (0.420)
Pos. surprised (within above arm)	0.319 (0.771)			−0.496 (1.721)		
Neg. surprised (within below arm)		0.332 (0.535)			0.449 (0.962)	
Negatively reassured			−0.332 (0.535)			−0.449 (0.962)
Positively surprised			−0.314 (0.775)			0.658 (1.736)
Positively reassured			−0.633** (0.308)			1.155** (0.549)
No feedback – optimistic			−0.397 (0.313)			0.499 (0.532)
No feedback – pessimistic			−0.371 (0.593)			1.464** (0.638)
N	47	48	145	47	48	145
R^2	0.005	0.010	0.029	0.004	0.006	0.055

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS with HC3-robust SEs in parentheses. Cols. (1) and (4): above group only ($N = 47$); positively surprised = told above but predicted below ($n_{\text{surp}} = 5$). Cols. (2) and (5): below group only ($N = 48$); negatively surprised = told below but predicted above ($n_{\text{surp}} = 37$). Constant in cols. (1)–(2) and (4)–(5) = mean of the not-surprised sub-group within each arm. Cols. (3) and (6): full sample ($N = 145$); baseline = negatively surprised. Negatively reassured = told below, predicted below ($n = 11$); positively surprised = told above, predicted below ($n = 5$); positively reassured = told above, predicted above ($n = 42$); no feedback – optimistic = no feedback, predicted above ($n = 43$); no feedback – pessimistic = no feedback, predicted below ($n = 7$). Constant in cols. (3) and (6) = mean of the negatively surprised group. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

attribution and redistribute \$0.63 less than those whose expectation was contradicted (*negatively surprised*), both significant at the 5 percent level. This finding is consistent with motivated reasoning: comparative feedback amplifies the self-serving attribution effect most strongly when it aligns with the recipient’s prior self-view. Near-universal overconfidence (84% predicted above) limits the statistical power for the within-arm comparisons and prevents estimation of the symmetric case, but the asymmetry in magnitudes points toward a pattern in which positive self-views are reinforced more readily than revised downward.

Taken together, these findings support a self-serving attribution account of redistribution preferences: comparative performance signals shift merit-oriented beliefs, which in turn shift redistribution; and this shift is strongest when feedback *confirms* rather than challenges an existing self-view. The broader implication is that relative performance information is not a neutral signal — it interacts with prior beliefs in ways that can entrench meritocratic worldviews and reduce redistribution support, particularly among those who consistently perceive themselves as high performers. In a world where performance rankings are increasingly visible — on professional platforms, in academic settings, and on social networks — the self-serving dynamics documented here could contribute to a systematic polarisation of redistribution preferences between those who “win” and those who “lose” such comparisons.

Future work should (i) increase sample size to improve power for the individual contrasts with the no-feedback anchor and for the surprise/reassurance decomposition; (ii) test whether the effect persists when participants receive absolute score information alongside relative feedback, to disentangle comparative and informational channels; and (iii) recruit samples with greater variation in prior expectations, enabling clean identification of all six feedback types — including confirmed pessimists and surprised optimists — to fully characterise the interaction between prior beliefs and feedback valence.

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Appendix

Table A2: Treatment Effects on Effort Attribution and Redistribution
(Baseline: No Feedback)

	USD Redistributed		Effort Beliefs	
	(1)	(2)	(3)	(4)
Constant	2.120*** (0.199)	3.633*** (0.813)	6.956*** (0.290)	5.994*** (1.180)
Told: Above	−0.205 (0.281)	−0.172 (0.290)	0.467 (0.453)	0.476 (0.450)
Told: Below	0.318 (0.283)	0.377 (0.293)	−0.737 (0.471)	−0.825* (0.470)
Age		−0.008 (0.012)		0.021 (0.018)
Female		−0.256 (0.237)		0.324 (0.405)
Colombian		−0.402 (0.296)		0.413 (0.533)
Task score		−0.005 (0.006)		−0.007 (0.009)
Bachelor's degree		−0.363 (0.365)		0.369 (0.549)
Master's or above		−0.372 (0.404)		0.461 (0.597)
<i>N</i>	145	145	145	145
<i>R</i> ²	0.024	0.077	0.044	0.110
Demographic controls	No	Yes	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS with HC3-robust SEs in parentheses. Baseline: No Feedback. Cols. (1)–(2): USD redistributed to lower-earning worker (merit scenario). Cols. (3)–(4): effort–luck belief (0 = luck, 10 = effort). Controls: age, female indicator, Colombian indicator, task score, and education (Less than bachelor = baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Mediation Analysis (Baseline: No Feedback)

	(1) Merit redistrib. (\$)	(2) Merit redistrib. (\$)
Constant	3.542*** (0.424)	4.796*** (0.768)
Told: Above	-0.110 (0.274)	-0.080 (0.280)
Told: Below	0.167 (0.272)	0.217 (0.281)
Effort-luck belief	-0.204*** (0.051)	-0.194*** (0.053)
Age		-0.004 (0.011)
Female		-0.193 (0.229)
Colombian		-0.321 (0.291)
Task score		-0.007 (0.005)
Bachelor's degree		-0.292 (0.339)
Master's or above		-0.283 (0.378)
N	145	145
R^2	0.140	0.174
Demographic controls	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS with HC3-robust SEs in parentheses. Dependent variable: USD redistributed (merit scenario). Includes effort-luck belief as control to assess mediation; compare treatment coefficients with Table A2 cols. (1)–(2). Baseline: No Feedback. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Surprise Effects: Within-Arm and Full-Sample Comparisons (Baseline: No Feedback – Optimistic)

	USD Redistributed			Effort Beliefs		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	1.881*** (0.210)	2.182*** (0.486)	2.116*** (0.219)	7.476*** (0.353)	5.873*** (0.865)	6.821*** (0.327)
Pos. surprised (within above arm)	0.319 (0.771)			-0.496 (1.721)		
Neg. surprised (within below arm)		0.332 (0.535)			0.449 (0.962)	
No feedback – pessimistic			0.027 (0.591)			0.965* (0.581)
Negatively surprised			0.397 (0.313)			-0.499 (0.532)
Negatively reassured			0.066 (0.533)			-0.948 (0.925)
Positively surprised			0.084 (0.773)			0.159 (1.716)
Positively reassured			-0.235 (0.304)			0.655 (0.481)
N	47	48	145	47	48	145
R^2	0.005	0.010	0.029	0.004	0.006	0.055

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS with HC3-robust SEs in parentheses. Cols. (1) and (4): above group only ($N = 47$); positively surprised = told above but predicted below ($n_{\text{surp}} = 5$). Cols. (2) and (5): below group only ($N = 48$); negatively surprised = told below but predicted above ($n_{\text{surp}} = 37$). Cols. (3) and (6): full sample ($N = 145$); baseline = no feedback – optimistic (no feedback, predicted above; $n = 43$). No feedback – pessimistic = no feedback, predicted below ($n = 7$); negatively surprised = told below, predicted above ($n = 37$); negatively reassured = told below, predicted below ($n = 11$); positively surprised = told above, predicted below ($n = 5$); positively reassured = told above, predicted above ($n = 42$). Constant in cols. (3) and (6) = mean of the no-feedback optimistic group. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Treatment Effects on Effort Attribution and Redistribution
(No Intercept)

	USD Redistributed		Effort Beliefs	
	(1)	(2)	(3)	(4)
Told: Above	1.915*** (0.199)	3.461*** (0.837)	7.423*** (0.348)	6.470*** (1.301)
No Feedback	2.120*** (0.198)	3.633*** (0.731)	6.956*** (0.313)	5.994*** (1.220)
Told: Below	2.438*** (0.211)	4.010*** (0.778)	6.218*** (0.354)	5.169*** (1.269)
Age		-0.008 (0.012)		0.021 (0.018)
Female		-0.256 (0.237)		0.324 (0.405)
Colombian		-0.402 (0.296)		0.413 (0.533)
Task score		-0.005 (0.006)		-0.007 (0.009)
Bachelor's degree		-0.363 (0.365)		0.369 (0.549)
Master's or above		-0.372 (0.404)		0.461 (0.597)
<i>N</i>	145	145	145	145
<i>R</i> ²	0.024	0.077	0.044	0.110
Demographic controls	No	Yes	No	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes. OLS without intercept (-1) with HC3-robust SEs in parentheses. Each treatment coefficient equals the conditional group mean. Cols. (1)–(2): USD redistributed to lower-earning worker (merit scenario). Cols. (3)–(4): effort–luck belief (0 = luck, 10 = effort). Controls: age, female indicator, Colombian indicator, task score, and education (Less than bachelor = baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.